

MUHAMMAD HASEEB IMRAN

MECHANICAL & CAD DESIGN ENGINEER | AUTOMOTIVE & PRODUCT SPECIALIST

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🌐 LinkedIn Profile

PROFESSIONAL PROFILE

Results-driven **Mechanical & CAD Design Engineer** with extensive expertise in 3D parametric modeling, multi-body kinematics, Class-A surfacing, and **Design for Manufacturing & Assembly (DFM/DFA)**. Proven track record of architecting production-ready powertrain systems, precision sheet-metal enclosures, compliant fluid mechanisms, and 6-DOF robotic manipulators. Highly adept at translating complex engineering requirements into high-tolerance fabrication models, ISO/ASME Y14.5 GD&T drawings, and validated FEA/CFD simulations.

FLAGSHIP ENGINEERING PROJECTS

High-Compression V8 Powertrain Assembly (184 Components) **Lead CAD Engineer**

Powertrain & Kinematics | SolidWorks, Motion Simulation, EASM

- Architected a complete **184-part V8 engine assembly** from bottom-up: crankshaft, connecting rods, pistons, camshafts, and valvetrain synchronization.
- Executed **dynamic interference checking and motion studies** to eliminate part collisions during 360° rotational cycles.
- Generated high-resolution exploded-view assembly kinematics for manufacturing documentation and service sequence planning.

6-DOF Industrial Robotic Manipulator (680mm Reach) **Mechanical Designer**

Robotics & Automation | SolidWorks, STEP, FEA Structural Analysis

- Engineered a 6-axis articulated robotic arm featuring internal actuator housings, harmonic drive mounts, and integrated cable management.
- Conducted **finite element analysis (FEA)** on structural arm links under full-extension static and dynamic payload conditions.
- Standardized precision bearing seats and CNC-machined aluminum linkages adhering to tight ISO tolerances.

Modular Industrial Reduction Gearbox (16:1 Ratio) **Transmission CAD Modeler**

Power Transmission | SolidWorks, Gear Train Synthesis, DFM

- Developed a high-torque **16:1 multi-stage reduction gearbox (67 parts)** with aligned shaft geometry and inspection-first casing design.
- Calculated gear tooth bending/contact stresses and designed custom oil-bath lubrication fluid channels.
- Integrated split-housing architecture for rapid disassembly, bearing replacement, and field serviceability.

CORE COMPETENCIES

3D CAD & MODELING

SolidWorks CATIA V5

Fusion 360 AutoCAD

Parametric Modeling

Class-A Surfacing

Multi-Body Assemblies

MANUFACTURING & DFM

ASME Y14.5 GD&T DFM / DFA

Sheet Metal Fabrication

K-Factor & Bends

Plastic Injection Molding

CNC Machining (3/5-Axis)

Die-Casting Tooling

3D Rapid Prototyping

ANALYSIS & SIMULATION

ANSYS FEA

Stress & Strain Analysis

Thermal Dissipation

CFD Flow Paths

Motion Simulation

Tolerance Stack-up

DOCUMENTATION & TOOLS

2D Technical Drafting

BOM Generation

STEP / IGES / STL

EASM Exploded Models

Engineering Blueprinting

EDUCATION

Compliant Diaphragm Water Pump (12 L/min Flow)

Fluid Systems Engineer

Fluid Mechanics | SolidWorks, 2D Drafting (PDF), CFD Flow Validation

- Engineered a 42-part fluid displacement pump utilizing an eccentric cam drive and compliant elastomer diaphragm.
- Generated comprehensive 2D manufacturing blueprints with GD&T for injection molding and sealing O-rings.
- Streamlined flow chamber geometries via fluid path analysis to minimize cavitation and maximize delivery pressure.

Precision Sheet Metal Power Supply Enclosure

Sheet Metal Specialist

Sheet Metal & Thermal | SolidWorks Sheet Metal, Flat Patterns, SLDPRT

- Designed a folded 1.5mm steel enclosure (18 bends) utilizing precise K-factor and bend deduction calculations for CNC press brake.
- Integrated tool-less snap-fit cover mechanisms, punched ventilation louvers, and EMI shielding boundaries.

Ergonomic Gaming Mouse Housing (82g Ultralight)

Industrial Product Designer

Consumer Electronics | Class-A Surfacing, Injection Molding DFM, STEP

- Formed complex ergonomic top-shell geometries using continuous curvature (G2/G3) surfacing tools.
- Engineered internal rib structures, screw bosses, and draft angles (1.5°–3°) tailored for high-volume plastic injection molding.

BS in Mechanical Engineering

Major in Machine Design & CAD/CAM

2021 – 2025

CERTIFICATIONS

- ✓ Certified SolidWorks Professional (CSWP) — Advanced Assembly & Surfacing
- ✓ ASME Y14.5 GD&T Certification — Geometric Dimensioning & Tolerancing
- ✓ Design for Manufacturing (DFM) — Tooling & Sheet Metal Specialist
- ✓ Finite Element Analysis (FEA) — Structural & Thermal Simulation Mastery

KEY STRENGTHS

- Engineering Rigor
- Cross-functional Design
- Design Optimization
- Rapid Iteration
- High Torque Kinematics